When Should an Electric Ride-Hailing Driver Charge?

Joint intra-day optimisation of charging time and order acceptance

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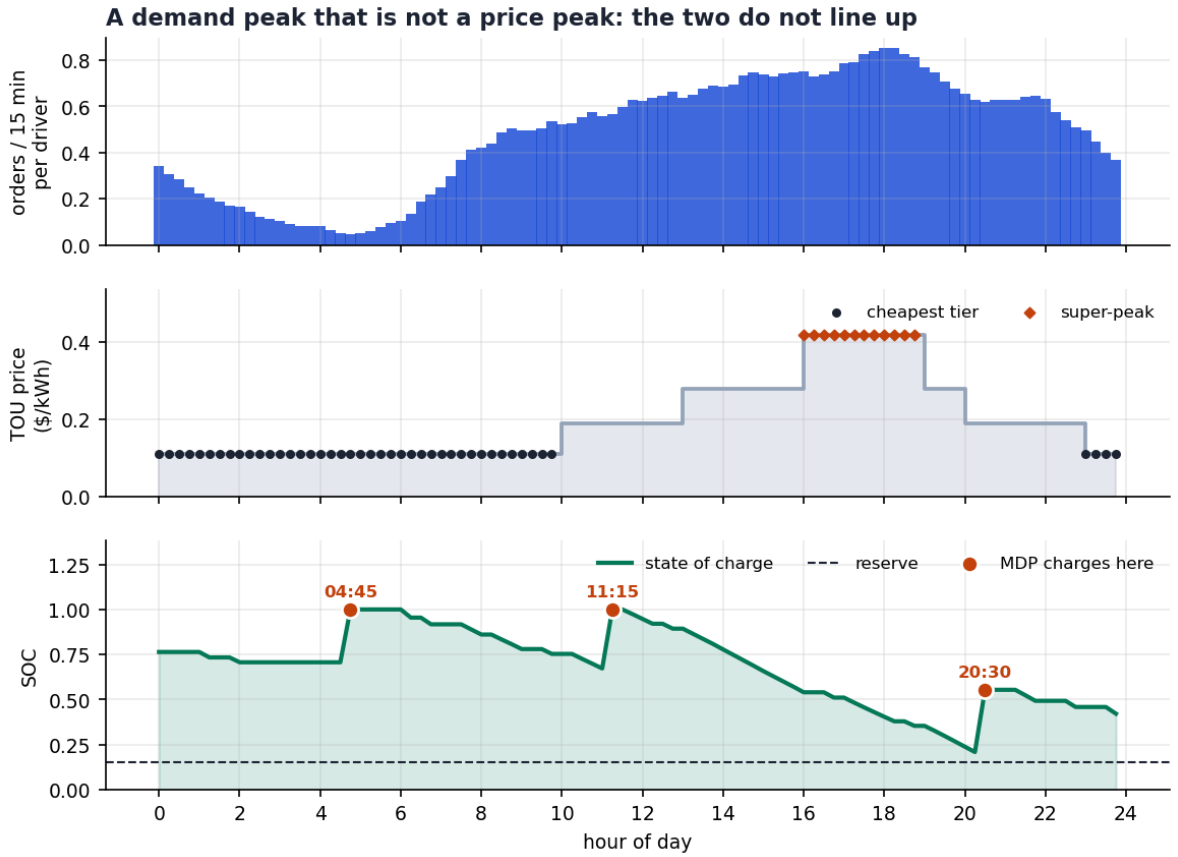
Course: Optimization Methods (084100551) · Supervisor: Prof. Zhenhong Lin

In one sentence: we model a single electric ride-hailing driver's charging and order decisions as a Markov decision process and show that the cheapest electricity hour is almost never the right hour to charge.

Background: the driver is paid to drive, and billed for the plug

Electric ride-hailing drivers face a time-of-use electricity tariff and an order flow that both vary over the day — but they do not vary together.

- A driver earns from serving orders; electricity is a pure cost with a time-of-use (TOU) tariff that is ~3.8× cheaper at night than at the 16:00–19:00 super-peak.
- Charging takes a 15-minute slot, and the driver keeps earning while the car charges — but only if orders are still arriving.
  - Opportunity cost, not the tariff, decides the answer: a slot that saves $0.31/kWh but sits in a dead hour is a bad slot.
- Naive rule ("charge in the cheapest hour") ignores this entirely, and it is exactly what real drivers and naive dispatchers propose.



One simulated day: demand (blue), tariff (grey), and the MDP charging policy (orange dots) on the state-of-charge curve.

Research question and contributions

A deliberately small, sharply-posed question — a single driver, one day, one battery.

> **RQ. When should an electric ride-hailing driver charge, and how much do we lose by charging in the cheapest hour instead of the optimal one?**

- **Contributions**
- A joint MDP of order acceptance and charging for a single driver: 8,256 states, 96 slots, Poisson arrivals summed analytically.
- A decomposition that shows exactly why the optimal policy departs from the tariff-following rule — the opportunity cost of an idle hour.
- A closed-form extension to the multi-day cycle, giving the shadow price of stored energy (μ = \$0.2414/kWh).
- Full reproducibility: public data, published seeds, and every number in the paper regenerated by one script.

The optimization problem

Decision variables, one objective, three constraints — and the trade-off that makes it worth optimising.

Decision variables

- $a(t) \in \{0, 1\}$: charge or not in slot t
- $u(t) \geq 0$: number of orders accepted in slot t
- $s(t)$: state of charge at the start of slot t
- π: the policy mapping $(t, s) \rightarrow (a, u)$

Constraints

- SOC reserve floor: $s(t) \geq 15\%$ at all times
- Energy balance: $s(t+1) = s(t) + \eta \cdot a(t) \cdot P \cdot \tau(s) \cdot \Delta t - d(t) \cdot u(t)$
- Capacity and slot limits: $s(t) \leq 100\%$, one slot per driver

Objective (max)

$$\max \; E\left[\Sigma_t \left(\text{fare}(t) \cdot u(t) - \text{price}(t) \cdot P \cdot a(t) \cdot \Delta t \right) \right]$$

earnings from accepted orders, minus the electricity bill

The key trade-off

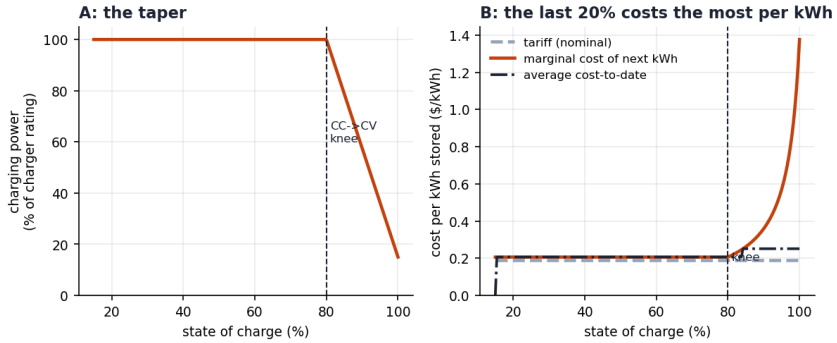
Every slot spent charging is a slot not spent earning — but the value of that slot depends on the demand profile, not on the tariff.

Cheap electricity + no orders = a loss. Expensive electricity + peak orders = a smaller loss.

Data and assumptions

One public dataset for demand; the tariff and vehicle parameters are stated assumptions, not measurements — and we say so explicitly.

Parameter	Value	Source / note
Order arrival rate λ(t)	91 → 1,684 orders/h	NYC TLC trip records, 2024-01
Fare rate	$53.0/h mean, $93.5/h peak	derived from trip fare ÷ trip duration
Battery / charger	40 kWh, 60 kW, η = 92%	typical BEV ride-hailing vehicle
Reserve floor / knee	15% SOC / 80% SOC	CC–CV taper above the knee
TOU tariff	$0.11 – $0.42 / kWh	assumption; 4 tiers, 44 valley slots
Horizon / step	96 slots × 15 min = 24 h	a full operating day



CC–CV taper and the resulting cost per kWh stored.

- What is measured and what is assumed is the honesty line of this paper: the demand profile is measured, the tariff is an assumption, and the sensitivity analysis in Section 7 varies the assumption.

Method: three models, each answering one question

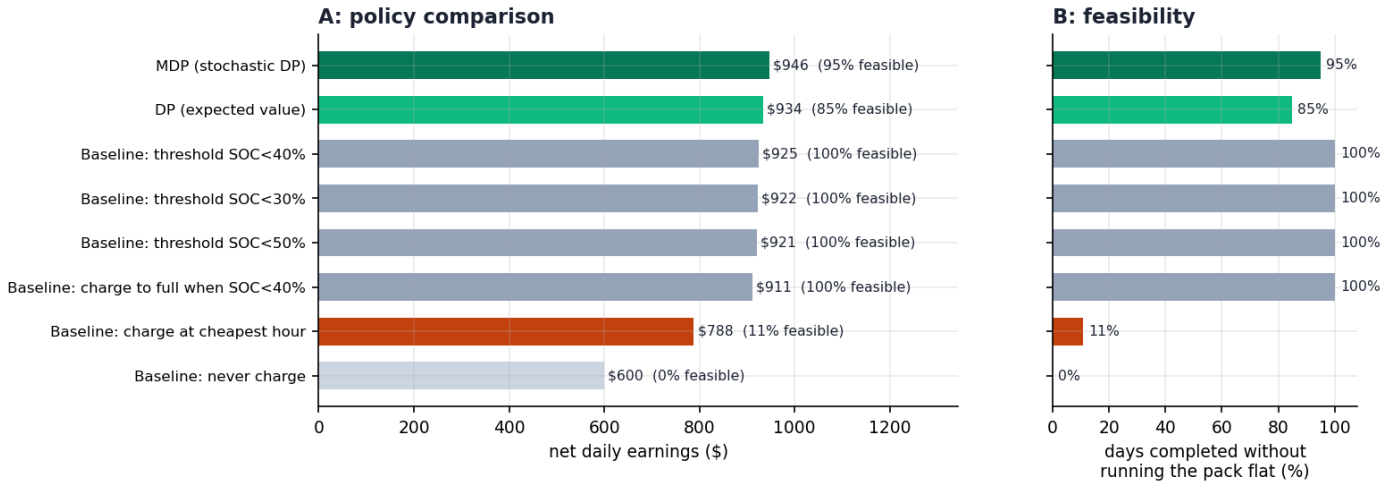
Each step is justified by what the previous one could not resolve.

Model	Question it answers	Why this method
DP expected value	What is the best schedule if the future is known in advance?	A lower bound and a sanity check; deterministic and easy to verify by hand
MDP Poisson arrivals	What should the driver do given only the current state?	Orders arrive stochastically; the Bellman recursion sums over the Poisson arrivals in closed form
Cyclic multi-day MDP	What is worth carrying into tomorrow?	The terminal value is the value of starting tomorrow, so it is a fixed point, solved by value iteration

- **Why not MILP?**
- A MILP is possible, but the stochastic arrival process makes the objective non-linear in the decision variables, and we would have to sample scenarios — trading a 20-minute solve for 50 approximation error.
- The MDP is exact and its structure is interpretable: the policy can be read off slot by slot, which is what makes the key insight visible at all.
  - No code on the slides: the Bellman recursion is the whole method — V(t, s) = max over actions of [immediate reward + expected continuation].

Baseline and comparison: what does optimisation buy?

Eight policies, 570 simulated feasible days, same demand realisation for every policy.



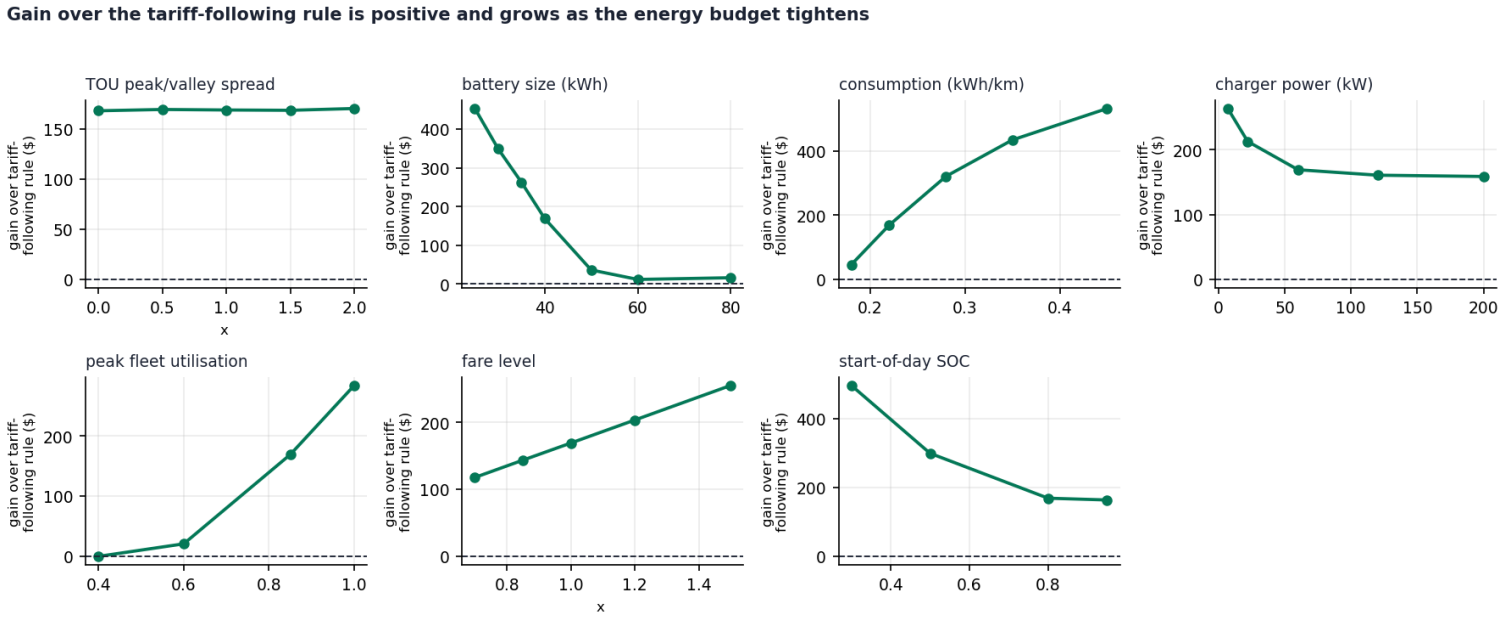
Net daily earnings and feasibility rate of the eight policies.

Policy	Net ($/day)	Feasible
MDP (ours)	$946.46	95%
DP (perfect forecast)	$934.15	85%
Charge-to-full rule	$911.06	82%
Cheapest-hour rule	$788.21	11%
Never charge	$600.09	0%

- **+$158.25/day (+20.1%) over the cheapest-hour rule**
- **+$21.47/day (+2.3%) over a perfect-forecast deterministic DP**
- The perfect-forecast DP is not the benchmark to beat — it wins on earnings but fails feasibility 10% more often, because a static plan cannot adapt to a bad draw.
- Feasibility is a first-class metric: a policy that strands the driver at 0% SOC is worthless even at a high average.

Sensitivity and extension

Does the conclusion survive when the assumptions move?



Gain of the optimal policy over the tariff-following rule, 7 one-factor sweeps.

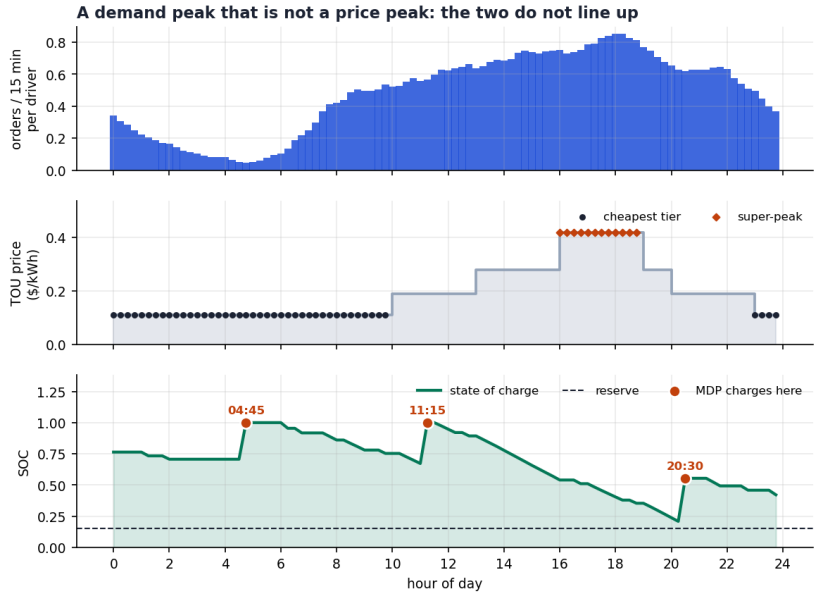
- 7 sweeps: TOU spread, battery size, consumption, charger power, peak utilisation, fare level, start-of-day SOC
- The gain is positive everywhere and grows as the energy budget tightens — optimisation matters more when energy is scarcer.
- **Ablations A1–A6 isolate every mechanism**
- Removing the opportunity-cost term from the reward is the only ablation that changes the policy qualitatively — and it turns the MDP back into the cheapest-hour rule.
- **Multi-day cycle**
- Cyclic MDP: μ = $0.2414/kWh shadow price, optimal start-of-day SOC 36.8%, steady-state charging at 04:45 and 12:00.

Key insight: the cheapest hour is the wrong hour

This is the one figure we would keep if we could only keep one.

> The tariff spread is $0.31/kWh, but the demand spread over the same 24 hours is 18.5×. Opportunity cost dominates, so the optimal policy buys cheap energy in slots where it is cheap — and charges where the demand gap is small.

- **Measured on the optimal policy:**
- 74.4% of feasible days contain at least one off-valley charge; 25.6% are valley-only.
- 100% of days charge in the deep night (00:00–05:00).
- Modal slot: 04:45 (87.2% of days); then 20:30 (9.5%), 12:30 (9.0%).
- **Why 20:30 and 12:30 appear:**
- They are shoulder-tier (2× cheaper than peak) AND still inside the demand shoulder — the only slots that satisfy both conditions.
- **The general lesson:**
- Price a time slot by its opportunity cost, not by its sticker price.



The optimal day: charge at the shoulder boundary, not at the tariff minimum.

Conclusion

Direct answer to the research question, with the conditions attached.

- **Finding 1.**
- Charging at the tariff minimum is the worst reasonable rule: $788.21/day and 11% feasibility, against $946.46/day and 95% for the optimal policy — a 20.1% gain.
- **Finding 2.**
- The optimal policy is strongly valley-anchored (74.4% of days charge off-valley, 100% charge after midnight) but not valley-exclusive: 25.6% of days stay entirely in the valley.
- **Finding 3.**
- The decision is driven by demand availability, not by price: 18.5× demand spread vs a 3.8× tariff spread, so the opportunity cost of an idle slot sets the optimum.
- **Finding 4.**
- Across a multi-day cycle, stored energy is worth μ = $0.2414/kWh and the optimal start-of-day SOC is 36.8%.

Limitations and next steps

Single driver (the fleet-scale version needs a different treatment); the tariff is an assumption, not a measured price; revenue is modelled as a deterministic rate. Next: multi-agent congestion, and pricing the charging slot as a market.

Backup: verification and reproducibility

Because every number in the paper is regenerated by one script.

Check	Result
Planner vs simulator consistency	2.25 standard errors (consistent rule); 2.31 SE (legacy rule)
Bellman residual (cyclic MDP)	9.29×10^{-10} after 34 sweeps
Formula validation	105 LaTeX formulas checked with KaTeX, 0 failures
Paper equations in Word	101 native OMML equation objects, 0 pasted images
Data	2,803,713 cleaned trips; 2,739,936 in-profile over 30 full days
Reproduction	code/ + data/ + published seeds; one command regenerates all figures and tables

- **AI use statement (also in the paper):**

- Generative AI was used for literature understanding, model discussion, code debugging and English-language editing. All results, equations and code logic were verified by the team; the authors take full responsibility for the submitted content.

Backup: expected questions

The guide lists eight questions we should be ready for; three of them have non-trivial answers.

Question	Short answer
Why is this an optimization problem?	Two coupled decisions (when to charge, which orders to take) compete for the same scarce resource: the driver's time and battery.
Why is this constraint necessary?	Without the 15% reserve floor the DP would plan runs that strand the driver at 0% SOC; we proved the constraint is non-redundant and measured the resulting infeasibility.
Why this method?	The arrivals are stochastic and the state is (time, SOC) — exactly the structure of an MDP with a closed-form Poisson expectation.
Why does the optimal solution have this shape?	Because the value of a slot is the demand it forgoes; when demand is flat, the tariff decides, and when demand is steep, the tariff cannot outvote it.
What if a key parameter changes?	7 sensitivity sweeps: the gain over the tariff-following rule stays positive and grows as the energy budget tightens.
Would the conclusion hold under different assumptions?	Yes for the tariff shape and fare level (re-calibrated VOT still gives 16 of 96 slots below $15.20/h) — but a flat tariff would collapse the problem to a pure energy-balance question.